

ABSTRACT

Brain MRI helps in studying changes in brain structure, but detecting early neurodegenerative abnormalities is still difficult. In medical conditions such as Alzheimer's Disease and Mild Cognitive Impairment, the changes can be small and slow, so they may not be clearly visible in the early stage. Many deep learning methods need labelled disease data for training. This becomes difficult when medical labels are limited or when the model has to be checked on cases that were not seen before.

In this work, a 3D convolutional autoencoder is used for brain MRI anomaly detection. The model is trained only with healthy MRI volumes so that it learns the normal reconstruction pattern of the brain. During testing, the model reconstructs MRI volumes from Healthy, MCI, and AD groups. The difference between the original and reconstructed volume is calculated and used as the anomaly score. Since MCI and AD samples are not included during training, the method works as an unsupervised anomaly detector instead of a direct disease classifier. A threshold selected from healthy validation data is used to decide whether a test volume is normal or abnormal.

XAI techniques are included through reconstruction difference maps and Grad-CAM maps, which show the regions that influenced the abnormality result. Alzheimer's Disease and Mild Cognitive Impairment are used as benchmark neurodegenerative scenarios to evaluate the method. From the obtained results, the model was able to separate healthy and abnormal MRI volumes in the selected test set. This study shows that a 3D autoencoder with visual explanation can be useful for MRI-based abnormality screening and neuroimaging analysis.

Keywords: XAI, Unsupervised Learning, 3D Autoencoder, Brain MRI, Alzheimer's Disease, MCI, Reconstruction Error, Grad-CAM.

CHAPTER 1

INTRODUCTION

1.1 Background

Neurodegenerative disorders are a major healthcare concern because they gradually affect brain structure, memory, cognition, and overall quality of life. Alzheimer's Disease (AD) is one of the most familiar neurodegenerative conditions and is often linked with an earlier stage called as Mild Cognitive Impairment (MCI). In MCI, cognitive decline is visible, but the symptoms may not fully satisfy the clinical criteria for dementia. Therefore, early identification of structural brain changes is important for timely screening, continuous monitoring, disease management, and clinical decision support.

Magnetic resonance imaging (MRI) is widely used in neuroimaging because it provides structural information about brain anatomy. Publicly available datasets such as ADNI and OASIS have supported research on aging, MCI, and Alzheimer's Disease by providing structural MRI data for analysis [1], [2], [3]. Recently, deep learning methods have shown promising performance in MRI-based analysis of cognitively normal subjects, MCI cases, and AD cases [1], [2], [3]. Since brain MRI is naturally volumetric, three-dimensional models are preferred because they preserve spatial relationships across adjacent slices better than two-dimensional slice-based methods [2], [3], [4].

1.2 Problem Statement

Even though deep learning has achieved good performance in neurodegenerative MRI analysis, many existing approaches use supervised classification. These approaches require large amounts of labeled pathological data and are trained to classify predefined categories such as healthy, MCI, and AD. This creates limitations in practical clinical environments, where disease labels may be unavailable, incomplete, or uncertain. Another limitation is that supervised models are often disease-specific. A model trained to recognize one particular disease category struggles to identify unfamiliar abnormalities or MRI data collected from many clinical centers. In addition, subtle neurodegenerative changes related to MCI and early AD are hard to find because they are gradual, distributed across brain regions, and may overlap with normal aging patterns. Hence, there is a requirement for a framework that can detect abnormal neurodegenerative patterns without depending on disease-specific labels, while also providing interpretable visual evidence for clinical proof.

1.3 Objective

The main aim of this work is to develop an XAI-based 3D unsupervised framework for early neurodegenerative anomaly detection in brain MRI.

The specific objectives are:

- i. To train a 3D convolutional autoencoder using only healthy MRI volumes.
- ii. To learn normal brain anatomical representations without using pathological labels during training.
- iii. To detect abnormal MRI volumes using volumetric reconstruction error.
- iv. To apply a percentile-based threshold derived from healthy validation data for normal/abnormal decision-making.
- v. To improve interpretability using voxel-wise reconstruction difference maps and Grad-CAM-based volumetric attention visualization.
- vi. To test the model's efficacy against neurodegenerative benchmarks, specifically Alzheimer's disease and mild cognitive impairment.

1.4 Scope of the Study

This work focuses on unsupervised anomaly detection from volumetric brain MRI data. The proposed framework is trained only on healthy/cognitively normal MRI volumes and tested on healthy, MCI, and AD MRI volumes. The aim is not to perform multi-class disease classification, but to identify whether an MRI volume deviates from the learned healthy brain pattern.

The scope of the study includes 3D MRI preprocessing, healthy-only training, 3D convolutional autoencoder-based reconstruction, reconstruction error-based anomaly scoring, threshold-based abnormality detection, and XAI-based visual interpretation. The framework is designed as a screening-support and clinical decision-support approach rather than a replacement for expert medical diagnosis.

1.5 Overview of the Proposed Approach

The proposed framework begins with volumetric MRI data used from benchmark datasets such as ADNI and OASIS. After pre-processing and normalization, the 3D convolutional autoencoder is trained using healthy MRI volumes only. The trained model learns the normal anatomical structure of the brain and reconstructs input MRI volumes. During the test, the trained model is fed with MRI volumes of healthy, MCI and AD subjects. The reconstruction error is the difference between the original and reconstructed MRI volume .

The larger the reconstruction error, the more the input volume deviates from the learned healthy anatomical pattern and therefore can be considered abnormal. Normal or abnormal test volumes are determined by using a threshold derived from healthy validation data. The framework provides XAI-based outputs like voxel-wise reconstruction difference maps and Grad-CAM-based attention maps to increase the interpretability of the system. These visualizations help to illustrate the regions contributing to the anomaly decision, making the model more transparent and interpretable for clinical understanding.

1.6 Main Contributions

The main contributions of this paper are as follows:

- i. A 3D unsupervised brain MRI anomaly detection framework trained exclusively on healthy volumetric MRI data.
- ii. A disease-agnostic anomaly scoring method based on volumetric reconstruction error and healthy-validation thresholding.
- iii. Integration of XAI visualizations, including voxel-wise reconstruction difference maps and Grad-CAM-based volumetric attention maps.
- iv. Evaluation using Alzheimer's Disease and Mild Cognitive Impairment as challenging neurodegenerative benchmark scenarios.
- v. A clinically interpretable framework that supports AI-assisted neuroimaging screening without relying on disease-specific labels during training.

CHAPTER 2

LITERATURE SURVEY

2.1 Background of Neuroimaging-Based Analysis

Deep learning has grown into a major area of study for the detection and screening of neurodegenerative disorders such as Alzheimer's Disease and Mild Cognitive Impairment. Structural MRI is useful in this area because it provides anatomical information related to brain atrophy, tissue degeneration, and disease progression.

Recent MRI-based studies have supported research on normal aging, MCI, and Alzheimer's Disease using structural imaging data. These studies are often used as benchmark references for evaluating machine learning and deep learning models in neurodegenerative MRI analysis [1], [2], [3].

2.2 Deep Learning for AD and MCI Detection

Various researchers have applied supervised deep learning methods to classify Alzheimer's Disease, Mild Cognitive Impairment, and cognitively normal subjects using brain MRI. Zarei et al. [1] proposed an MRI-based classification approach using radiomic features and using convolutional neural networks to distinguish AD, MCI, and cognitively normal participants. Their study shows the usefulness of CNN-based methods in neurodegenerative MRI analysis.

According to recent review research, deep learning models—particularly convolutional neural networks—have shown encouraging outcomes in Alzheimer's Disease detection from structural imaging data [2]. Other recent MRI-based classification studies have also highlighted the importance of deep learning for early Alzheimer's Disease detection [3]. These works show that supervised learning can produce good classification performance when sufficient labeled data is available.

However, supervised methods strongly depend on disease labels. This becomes a limitation in real clinical situations, where labels may be missing, uncertain, or difficult to obtain. In addition, supervised classifiers are usually trained for fixed disease types, which reduces their ability to handle unseen abnormalities or data from different clinical environments. Therefore, even though supervised deep learning approaches are useful, their dependence on labeled pathological data remains a major challenge for practical deployment.

2.3 3D MRI-Based Deep Learning Approaches

Brain MRI data is naturally volumetric, so 3D models are more effective than 2D slice-based models for preserving spatial anatomical information. Recent studies on 3D MRI-based Alzheimer's Disease analysis have shown that volumetric models are able to capture information across multiple brain slices instead of treating each slice separately [2], [3], [4].

Three-dimensional CNN models can preserve inter-slice continuity and capture structural patterns distributed across different brain regions. This is important because radiologists generally examine MRI scans as complete volumes rather than isolated 2D images.

Recent studies and reviews have also highlighted the importance of 3D imaging in Alzheimer's Disease analysis [2]. However, many 3D CNN-based approaches are still supervised and require disease-specific labels during training. This creates the need for 3D unsupervised frameworks that can learn normal brain anatomy without depending on pathological annotations.

2.4 Autoencoder Based Anomaly Detection

Unsupervised anomaly detection provides an alternative to supervised classification in medical imaging. Instead of learning disease labels, unsupervised models learn the normal anatomical pattern and detect anomalies as departures from this healthy, learned structure. Autoencoders are commonly used for this task because they compress the input image into a lower-dimensional representation and then reconstruct it. When trained on normal brain MRI data, an autoencoder is expected to reconstruct healthy structures accurately. However, when an abnormal MRI is given as input, the reconstruction error may increase because the abnormal pattern was not learned during training.

Recent studies have explored autoencoder-based and unsupervised approaches for detecting abnormalities in brain MRI without depending on disease-specific labels [4], [5], [6], [9]. These works support the idea of healthy-only training, where the model learns normal brain structure and identifies deviations during testing through reconstruction-based error analysis.

2.5 Recent Unsupervised and Disease-Agnostic Approaches

Recent unsupervised brain MRI anomaly detection studies have focused on learning normal brain patterns from unannotated or healthy data and detecting deviations during testing. Transformer-based masked autoencoder methods have also been explored for unsupervised

anomaly detection in brain MRI [5]. Similarly, recent disease-detection frameworks using unannotated T1-weighted brain MRI data have shown that abnormality detection can be performed without depending completely on disease-specific labels [6].

These studies support the idea that unsupervised models can detect deviations from normal anatomy without being restricted to a single disease class. This is useful for disease-agnostic abnormality screening, where the model identifies whether a scan differs from the learned healthy distribution rather than directly predicting a specific disease label [5], [6], [9].

However, many existing unsupervised anomaly detection works focus on obvious abnormalities such as lesions, tumors, or great structural changes. Detecting subtle neurodegenerative abnormalities such as MCI and early Alzheimer's Disease remains more difficult because these changes are gradual and may overlap with normal aging.

2.6 3D Autoencoders for Neurodegeneration

Recent research has started exploring unsupervised 3D autoencoder based methods for Alzheimer's-related structural analysis. Avci et al. [4] investigated the unsupervised analysis of Alzheimer's Disease signatures using 3D deformable autoencoders. This study is closely related to the present work because it shows that 3D autoencoder models can capture disease-related structural patterns without fully depending on supervised classification.

However, many existing unsupervised 3D approaches focus mainly on representation learning or abnormality localization. Limited attention is given to a complete clinically interpretable framework that combines healthy-only training, volumetric anomaly scoring, benchmark evaluation on subtle neurodegenerative conditions, and XAI-based visualization. This creates an opportunity to develop a 3D framework that is both unsupervised and explainable.

2.7 Explainable AI in Neuroimaging

Explainability is important in medical imaging because clinicians need visual and understandable evidence before trusting AI-assisted outputs. Recent XAI-based studies in Alzheimer's neuroimaging have discussed the importance of visual explanation methods for improving transparency in deep learning models [7], [8].

In MRI analysis, heatmap-based visualizations can help verify whether the model is focusing on clinically meaningful brain regions. In anomaly detection, reconstruction difference maps are also useful because they directly show voxel-level differences between

the original MRI and reconstructed MRI. Recent explainable AI frameworks for MRI-based classification have also shown that XAI can improve model interpretability in medical imaging tasks [7], [8].

Earlier XAI-based 2D MRI anomaly detection work has shown that reconstruction-based abnormality analysis and explanation maps can support Alzheimer's-related MRI screening [10]. However, 2D slice-based methods may not fully capture the spatial continuity of anatomical changes across MRI volumes. Therefore, extending explainable anomaly detection into a 3D volumetric setting is more suitable for clinically realistic neuroimaging analysis.

2.8 Research Gap

From the reviewed literature, the following research gaps can be identified:

- i. Many Alzheimer's Disease and MCI detection models are supervised and depend on disease-specific labels.
- ii. Several methods use 2D MRI slices, which may lose important volumetric spatial information.
- iii. Existing unsupervised anomaly detection studies often focus on general brain abnormalities or strong pathological changes.
- iv. Subtle neurodegenerative abnormalities such as MCI and early AD remain difficult to detect.
- v. Many models lack sufficient explainability, which limits clinical trust and interpretability.

To address these gaps, this study proposes an XAI-based 3D unsupervised framework trained only on healthy MRI volumes. The model detects abnormality using volumetric reconstruction error and provides visual explanations through voxel-wise reconstruction difference maps and Grad-CAM-based volumetric attention visualization. The framework is evaluated using Alzheimer's Disease and Mild Cognitive Impairment as challenging neurodegenerative benchmark scenarios.

CHAPTER 3

PROPOSED SYSTEM

3.1 Overview of the Proposed System

The proposed system is developed for early detection of neurodegenerative abnormalities from volumetric brain MRI data. It uses an unsupervised learning approach, where the model is trained only on MRI volumes from healthy/cognitively normal subjects. The system is designed to learn the normal anatomical structure of the brain and identify abnormal MRI volumes based on their deviation from the learned healthy pattern. The system's central model is a 3D convolutional autoencoder. The volumetric nature of MRI data makes the 3D model efficient in preserving spatial relationships between adjacent slices. During testing, the trained autoencoder reconstructs unseen MRI volumes and the anomaly score is computed based on the difference between the original and the reconstructed volume. The proposed system also incorporates explainable artificial intelligence techniques for interpretation. Voxel-wise reconstruction difference maps and Grad-CAM based attention visualizations are generated to identify the regions related to the abnormality decision. Thus, the system provides both detection output and visual explanation and the overall workflow of the proposed system is shown in Figure 3.1.

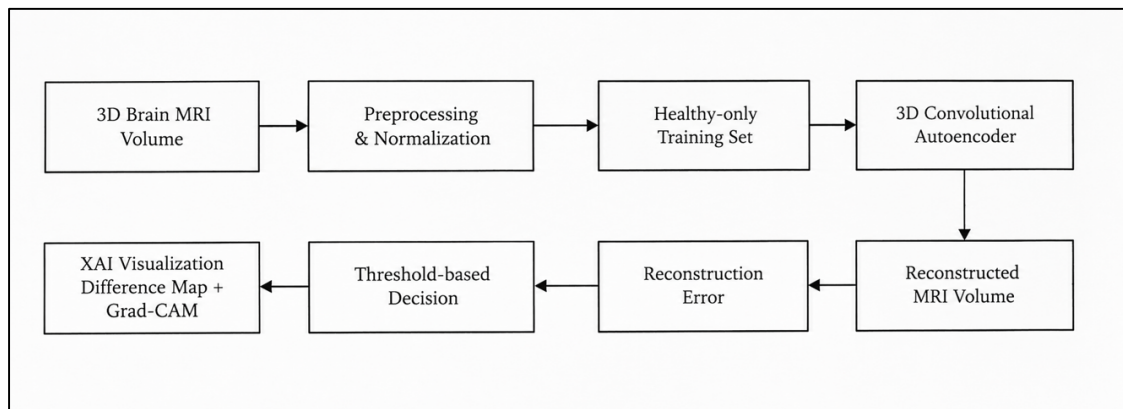


Figure 3.1 Overall workflow of the proposed XAI-based 3D unsupervised brain MRI anomaly detection framework.

3.2 Dataset Preparation

The proposed system uses publicly available structural MRI datasets such as ADNI and OASIS for neurodegenerative analysis. The dataset includes MRI volumes from

healthy/cognitively normal subjects, Mild Cognitive Impairment subjects, and Alzheimer's Disease subjects.

For model training, only healthy MRI volumes are used. MCI and AD MRI volumes are not included during training and are reserved only for testing and evaluation. This ensures that the model learns only normal brain anatomy and remains fully unsupervised during training.

The dataset is divided into three subsets:

Training set: Healthy MRI volumes only

Validation set: Healthy MRI volumes only

Testing set: Healthy, MCI, and AD MRI volumes

This arrangement as shown in Figure 3.2 allows the autoencoder to learn the normal anatomical representation during training and detect deviations in unseen abnormal cases during testing.

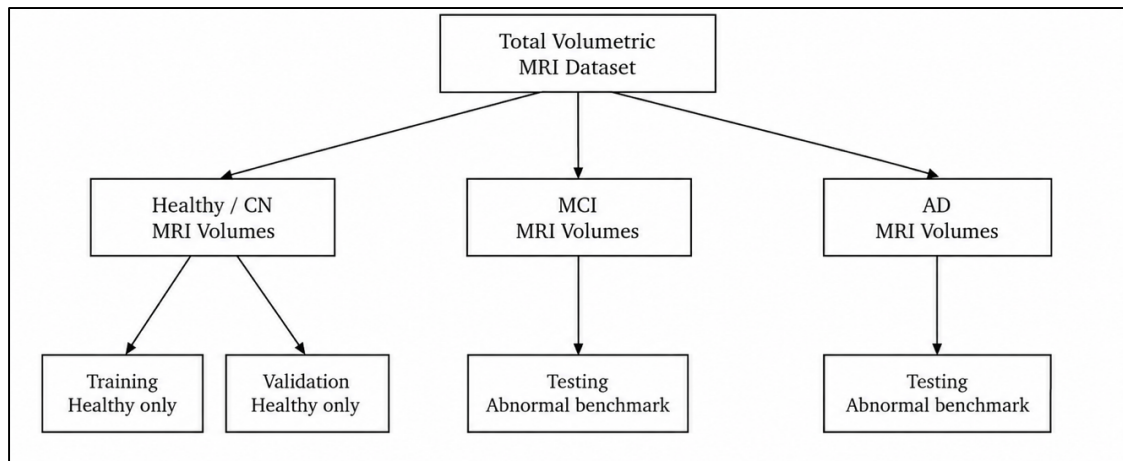


Figure 3.2 Dataset organization and experimental split used in the proposed framework.

3.3 MRI Preprocessing

MRI volumes collected from different subjects may vary in resolution, spatial size, intensity range, and orientation. Therefore, preprocessing is required to convert all MRI volumes into a standard format before giving them as input to the 3D convolutional autoencoder.

The preprocessing stage includes:

1. Loading the MRI volume
2. Selecting the required brain image data
3. Applying intensity normalization
4. Resizing the volume to a fixed spatial dimension
5. Adding the channel dimension

6. Preparing the volume for model input

As shown in Figure 3.3, each MRI scan is converted into a fixed-size 3D tensor of shape $64 \times 128 \times 128 \times 1$. Here, **64** represents the depth or number of slices, **128 × 128** represents the in-plane image resolution, and **1** represents the grayscale channel. Voxel intensity normalization is applied to bring values into the range $[0, 1]$, which improves training stability and reduces intensity variation between scans.

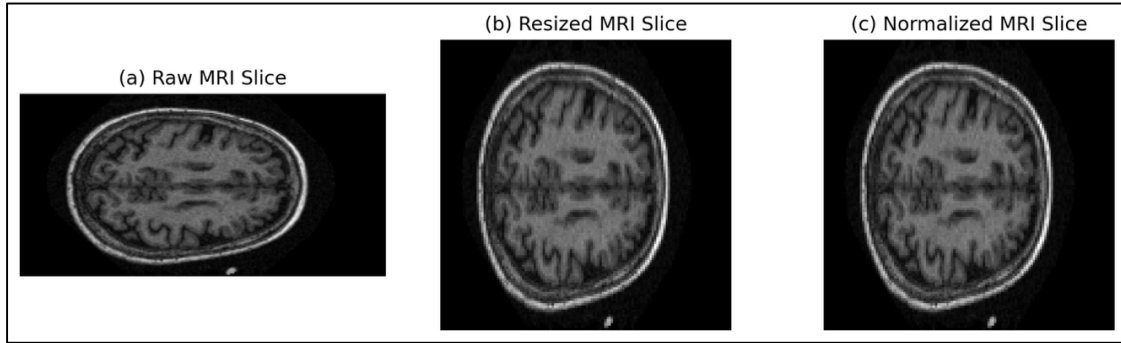


Figure 3.3 MRI preprocessing stages including raw MRI slice, resized slice, and normalized slice.

3.4 3D Convolutional Autoencoder Architecture

The main model used in the proposed system is a 3D convolutional autoencoder. The Architecture of the proposed 3D convolutional autoencoder is shown in Figure 3.4. It consists of two main parts: an encoder and a decoder. The encoder extracts relevant volumetric features from the input MRI and encodes them with a compact latent space. The decoder reconstructs the MRI volume from this latent representation. The encoder consists of Conv3D and MaxPooling3D layers for extracting spatial patterns in brain volume. The decoder consists of UpSampling3D and Conv3D layers for reconstructing input MRI volume. The model takes as input a 3D MRI volume and outputs a reconstructed MRI volume. During training the model is trained to minimize the difference between input and reconstructed output.

The reconstruction loss is calculated as Mean Squared Error.

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (X_i - \hat{X}_i)^2$$

where:

- X_i is the original voxel value,
- $\hat{X}_i$ is the reconstructed voxel value,
- N is the total number of voxels.

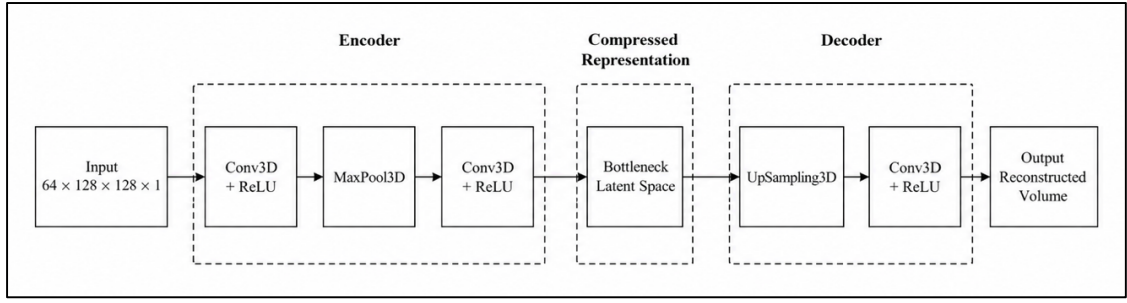


Figure 3.4 *Architecture of the proposed 3D convolutional autoencoder.*

3.5 Healthy-Only Training Strategy

The proposed system follows a healthy-only unsupervised training strategy. In this approach, only normal MRI volumes are used to train the 3D convolutional autoencoder. The model is not trained using MCI or AD MRI volumes.

The main idea is that a model trained only on healthy data should reconstruct healthy MRI volumes accurately. When an abnormal MRI volume is given during testing, the model may not reconstruct the abnormal regions properly because such patterns were not learned during training. As a result, abnormal volumes are expected to produce higher reconstruction errors.

This strategy avoids dependence on manually labeled pathological data and supports disease-agnostic anomaly detection.

3.6 Reconstruction-Based Anomaly Scoring

After training, each unseen test MRI volume is passed through the trained 3D autoencoder. The model reconstructs the input volume, and the difference between the original MRI and reconstructed MRI is calculated. This difference is used as the reconstruction-based anomaly score. A lower reconstruction error indicates that the input volume is close to the learned healthy brain pattern. A greater reconstruction error suggests that the volume deviates from normal anatomy and may be abnormal.

$$S = \frac{1}{DHW} \sum_{d=1}^D \sum_{h=1}^H \sum_{w=1}^W (X_{d,h,w} - \hat{X}_{d,h,w})^2$$

where:

- D = depth,
- H = height,
- W = width of the volume.

3.7 Threshold-Based Anomaly Detection

To classify MRI volumes as normal or abnormal, a threshold is calculated using reconstruction errors from healthy validation samples. The system employs a percentile thresholding approach. The threshold of anomaly is set to the 95th percentile of reconstruction errors on the healthy validation set. During testing, an anomaly score is assigned to each MRI volume. If the score is less than or equal to the threshold, the volume is a normal volume. If the score is higher than the threshold, the volume is marked as abnormal. The decision process is still unsupervised as the threshold is only calculated from healthy validation data. The anomaly threshold is computed as:

$$T = P_{95}(S_{healthy})$$

where P_{95} denotes the 95th percentile of anomaly scores from healthy validation samples.

The final decision rule is:

$$\text{Decision} = \begin{cases} \text{Normal}, & S \leq T \\ \text{Abnormal}, & S > T \end{cases}$$

3.8 XAI-Based Visual Interpretation

Explainability is important in medical imaging because the model should provide understandable evidence along with its prediction. Therefore, the proposed system includes XAI-based visual interpretation.

Two explanation methods are used:

1. **Voxel-wise reconstruction difference map:** This map is generated by calculating the absolute difference between the original MRI volume and the reconstructed MRI volume. It highlights regions where the model could not reconstruct the input accurately.
2. **Grad-CAM-based attention visualization:** Grad-CAM is used to highlight regions that contribute strongly to the anomaly-related response of the model. For 3D MRI, representative slices are selected from volumetric attention maps for visualization.

These visual outputs help improve transparency and allow users to observe the regions that influence the anomaly decision.

A voxel-wise difference map is obtained by computing the absolute difference between the original MRI volume and the reconstructed volume:

$$D_{map} = | X - \hat{X} |$$

This map highlights regions that the model could not reconstruct well, which may correspond to structurally abnormal areas.

3.9 Evaluation Strategy

The proposed system is evaluated using neurodegenerative benchmark scenarios involving healthy, MCI, and AD MRI volumes. The system is not designed as a direct multi-class disease classifier. Instead, healthy MRI volumes are treated as normal, while MCI and AD MRI volumes are treated as abnormal during evaluation.

The evaluation scenarios are:

- i. Healthy vs MCI
- ii. Healthy vs AD
- iii. Healthy vs MCI + AD

The performance is evaluated using accuracy, sensitivity, specificity, precision, F1-score, and Area Under the ROC Curve. Along with quantitative evaluation, qualitative analysis is performed using reconstruction maps and Grad-CAM-based visual explanations.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Hardware Requirements

The proposed system requires a computing environment that can support MRI preprocessing, autoencoder training, model evaluation, and output visualization. Since deep learning models require higher computational power, a GPU-supported environment such as Google Colab is preferred for faster execution and reduced local hardware dependency and to perform model training, evaluation, and output generation efficiently. The detailed hardware requirements needed for executing the proposed system are listed in Table 4.1.

Component	Requirement
Processor	Intel i5/i7 or equivalent processor
RAM	Minimum 8 GB, recommended 16 GB
GPU	Google Colab GPU or NVIDIA GPU support
Storage	Minimum 10 GB free space
System Type	64-bit operating system
Internet	Required for dataset access and Google Colab execution

Table 4.1 Hardware Requirements

4.2 Software Requirements

The implementation is carried out using Python and commonly used deep learning libraries. These tools are required for dataset handling, preprocessing, model development, training, evaluation, and visualization.

These software tools support the complete workflow of the proposed framework, including data loading, preprocessing, model training, performance evaluation, and visual explanation generation. This also remains consistent with the implementation tools used in the proposed framework, such as Google Colab, TensorFlow/Keras, NiBabel, NumPy, Scikit-learn, and Matplotlib. The software tools and libraries used for implementing the proposed system are listed in Table 4.2.

Software / Library	Purpose
Python	Main programming language
Google Colab	Cloud-based execution environment
TensorFlow / Keras	3D autoencoder model development and training
NumPy	Numerical and array-based operations
Pandas	Dataset and result organization
NiBabel	Loading and handling volumetric MRI files when NIFTI data is available
OpenCV / Scikit-image	MRI image preprocessing, resizing, and normalization
Scikit-learn	Performance evaluation metrics
Matplotlib	Graphs and visual output generation
Kaggle API	Dataset download and access for the current implementation

Table 4.2 Software Requirements

4.3 Dataset Requirements

The system requires brain MRI data grouped into healthy and neurodegenerative categories. In the submitted framework, public neuroimaging datasets such as ADNI and OASIS are considered for evaluation. For the current experimental implementation, an OASIS-based MRI dataset accessed through Kaggle is used to generate the required outputs while following the same healthy-only training strategy. The dataset requirements and category-wise usage in the proposed system are listed in Table 4.3.

Dataset Group	Data Category Used	Usage
Healthy / CN	Non Demented	Training, validation, and testing
MCI	Very Mild Demented	Testing only
AD	Mild Demented / Moderate Demented	Testing only
Labels	Folder-based class labels	Used only for evaluation
File Format	MRI image files arranged into 64-slice volumetric stacks	Used for preprocessing and model input

Table 4.3 Dataset Requirements and Usage

The training process uses only healthy MRI samples so that the model learns normal brain patterns. MCI and AD samples are used only during testing to evaluate whether the system can detect abnormal neurodegenerative changes using reconstruction error. This follows the healthy-only training strategy described in the proposed framework.

4.4 Functional Requirements

The system should perform the following functions:

1. Load MRI images from the dataset.
2. Organize the data into Healthy, MCI, and AD categories.
3. Preprocess MRI data by resizing and normalization.
4. Train the autoencoder using only healthy MRI samples.
5. Reconstruct test MRI samples.
6. Calculate reconstruction error as the anomaly score.
7. Classify samples as normal or abnormal using a threshold.
8. Generate evaluation outputs such as accuracy, precision, recall, F1-score, ROC curve, and confusion matrix.
9. Generate visual outputs such as reconstructed images, difference maps, and Grad-CAM-based heatmaps.

4.5 Non-Functional Requirements

The system should be efficient, reliable, and easy to execute in a research environment. It should generate stable outputs when the dataset is properly organized and preprocessed. The framework should also provide interpretable visual results so that detected abnormalities can be understood more clearly.

The framework can be extended to official OASIS and ADNI datasets when full dataset access is available, which can improve the strength and reliability of experimental validation. For the current implementation, Kaggle-hosted OASIS MRI data is used to generate the required outputs while maintaining the same workflow described in the submitted paper.

CHAPTER 5

IMPLEMENTATION

5.1 Experimental Environment

The implementation of the proposed framework is carried out using Python in a GPU-supported environment. Google Colab is used as the main development platform because it provides cloud-based execution, GPU support, and easy integration with deep learning libraries. This reduces local hardware dependency and allows the model training, testing, and output visualization to be performed efficiently.

The main tools used in the implementation include TensorFlow/Keras for building and training the autoencoder model, NumPy and Pandas for numerical and dataset-related operations, OpenCV/Scikit-image for image preprocessing, Scikit-learn for evaluation metrics, and Matplotlib for graph and visualization generation. Kaggle API is used to access the Kaggle-hosted OASIS MRI dataset for the current experimental implementation.

5.2 Dataset Usage

The dataset is organized into three main groups: healthy/cognitively normal, Mild Cognitive Impairment, and Alzheimer's Disease. In the current implementation, the Kaggle-hosted OASIS MRI dataset is used to generate experimental outputs while following the same healthy-only training strategy described in the proposed framework.

The healthy/non-demented samples are used for training, validation, and testing. The Very Mild Demented category is considered for MCI testing, while Mild Demented and Moderate Demented categories are considered for AD-related abnormal testing. The class labels are used only during evaluation and not during model training.

The dataset usage is arranged as follows:

Training set: Healthy / Non Demented MRI samples only

Validation set: Healthy / Non Demented MRI samples only

Testing set: Healthy, MCI, and AD MRI samples

This arrangement ensures that the model learns only normal brain patterns during training and detects abnormality based on reconstruction error during testing.

5.3 Preprocessing Implementation

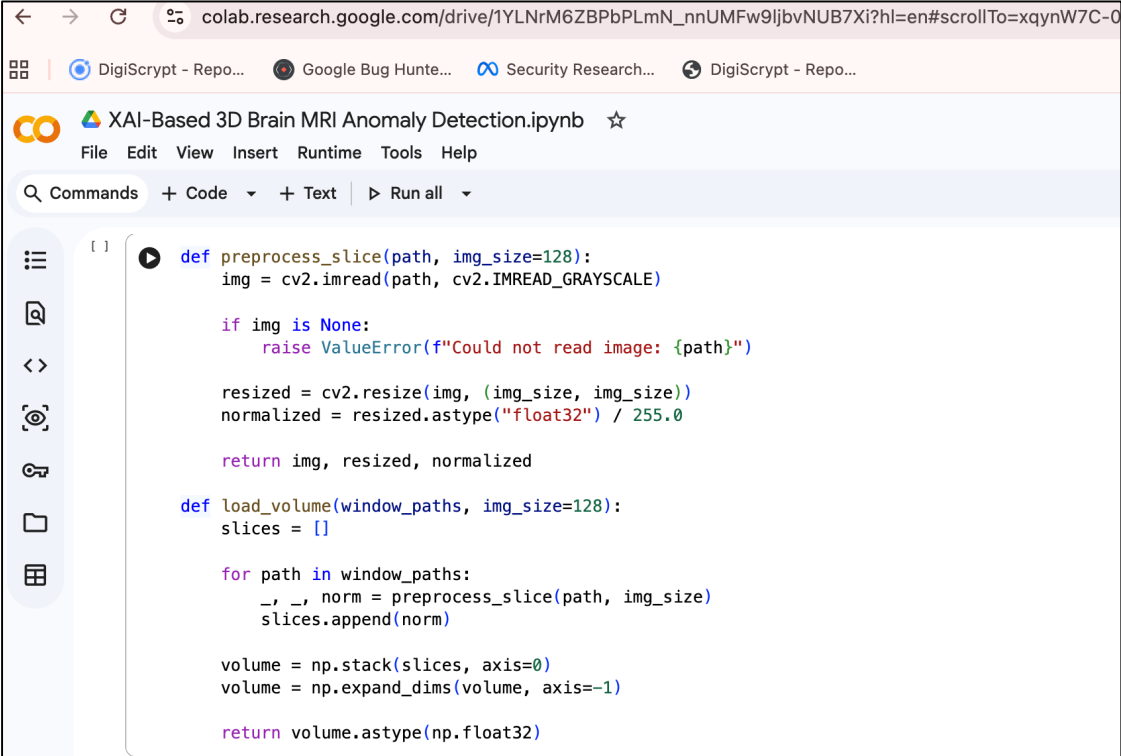
Before training the model, all MRI images are preprocessed to make them suitable for input into the autoencoder. The preprocessing step helps maintain uniform image size, intensity range, and input format.

The preprocessing process includes:

1. Loading MRI image files from the dataset folders.
2. Converting images into grayscale format if required.
3. Resizing each image or MRI slice to the required input dimension.
4. Normalizing pixel intensity values into the range **0 to 1**.
5. Expanding the channel dimension to match the model input format.
6. Splitting healthy samples into training and validation sets.

In the proposed framework, MRI volumes are represented using the input shape **64 × 128 × 128 × 1**. For the current implementation using Kaggle-hosted MRI image data, preprocessing is performed according to the available image format while maintaining the same reconstruction-based anomaly detection workflow.

The following code snippet in Figure 5.3 shows the preprocessing function used to resize MRI slices, normalize pixel values, and stack slices into a fixed-size 3D volume.



```
def preprocess_slice(path, img_size=128):
    img = cv2.imread(path, cv2.IMREAD_GRAYSCALE)

    if img is None:
        raise ValueError(f"Could not read image: {path}")

    resized = cv2.resize(img, (img_size, img_size))
    normalized = resized.astype("float32") / 255.0

    return img, resized, normalized

def load_volume(window_paths, img_size=128):
    slices = []

    for path in window_paths:
        _, _, norm = preprocess_slice(path, img_size)
        slices.append(norm)

    volume = np.stack(slices, axis=0)
    volume = np.expand_dims(volume, axis=-1)

    return volume.astype(np.float32)
```

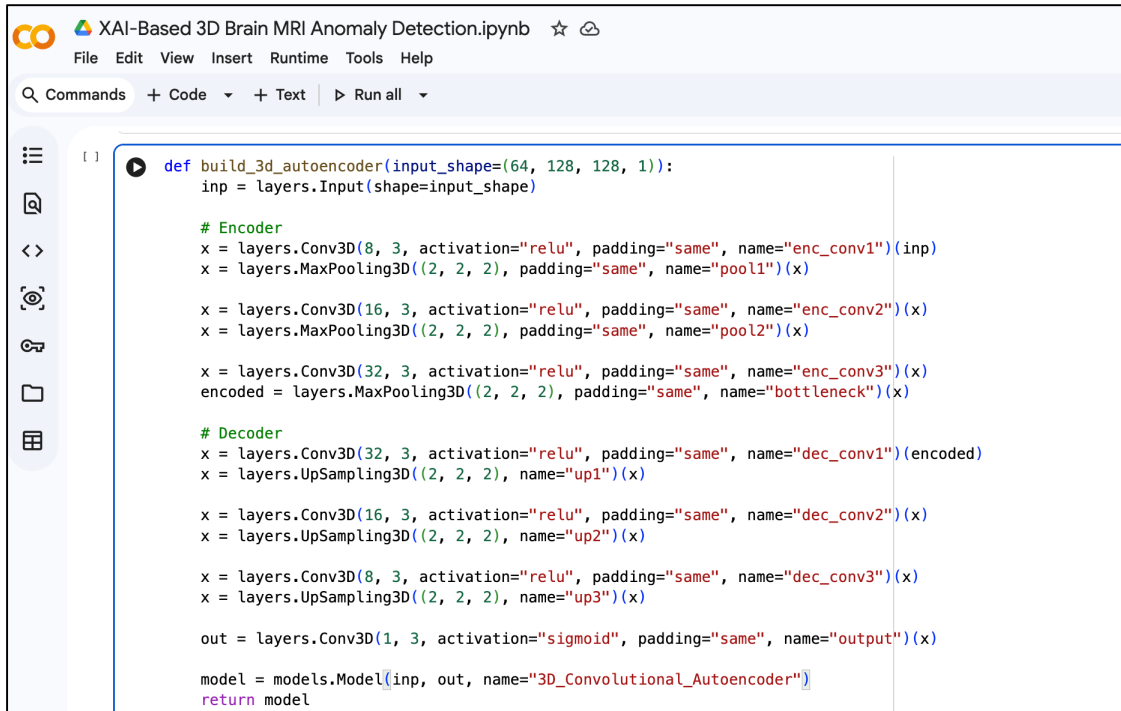
Figure 5.3 Code snippet for MRI preprocessing and 3D volume formation.

In this implementation, each MRI slice is converted into grayscale, resized to 128×128 , normalized to the range $[0, 1]$, and then stacked with other slices to form a 3D input volume. The final channel dimension is added so that the data matches the input format required by the 3D convolutional autoencoder.

5.4 Model Implementation

The main model used in this implementation is a 3D convolutional autoencoder. It is developed to reconstruct MRI volumes and support anomaly detection through reconstruction-based analysis. The model is trained only with healthy MRI volumes so that it can learn the normal reconstruction pattern of the brain.

The model was implemented using TensorFlow/Keras. The input was given in the form of a volumetric tensor with the size $64 \times 128 \times 128 \times 1$. The encoder part was created using Conv3D and MaxPooling3D layers to extract and compress spatial information from the MRI volume. The decoder part used UpSampling3D and Conv3D layers to reconstruct the input volume from the compressed representation. The model was compiled using the Adam optimizer and Mean Squared Error loss function. The implementation code is shown in Figure 5.4(a).



```
def build_3d_autoencoder(input_shape=(64, 128, 128, 1)):
    inp = layers.Input(shape=input_shape)

    # Encoder
    x = layers.Conv3D(8, 3, activation="relu", padding="same", name="enc_conv1")(inp)
    x = layers.MaxPooling3D((2, 2, 2), padding="same", name="pool1")(x)

    x = layers.Conv3D(16, 3, activation="relu", padding="same", name="enc_conv2")(x)
    x = layers.MaxPooling3D((2, 2, 2), padding="same", name="pool2")(x)

    x = layers.Conv3D(32, 3, activation="relu", padding="same", name="enc_conv3")(x)
    encoded = layers.MaxPooling3D((2, 2, 2), padding="same", name="bottleneck")(x)

    # Decoder
    x = layers.Conv3D(32, 3, activation="relu", padding="same", name="dec_conv1")(encoded)
    x = layers.UpSampling3D((2, 2, 2), name="up1")(x)

    x = layers.Conv3D(16, 3, activation="relu", padding="same", name="dec_conv2")(x)
    x = layers.UpSampling3D((2, 2, 2), name="up2")(x)

    x = layers.Conv3D(8, 3, activation="relu", padding="same", name="dec_conv3")(x)
    x = layers.UpSampling3D((2, 2, 2), name="up3")(x)

    out = layers.Conv3D(1, 3, activation="sigmoid", padding="same", name="output")(x)

    model = models.Model(inp, out, name="3D_Convolutional_Autoencoder")
    return model
```

Figure 5.4(a) Code snippet for implementing the 3D convolutional autoencoder model.

The Keras model summary was used to verify the implemented architecture. It shows the input shape, 3D convolution layers, pooling layers, upsampling layers, output shape, and trainable parameters of the model. The generated model summary is shown in Figure 5.4(b).

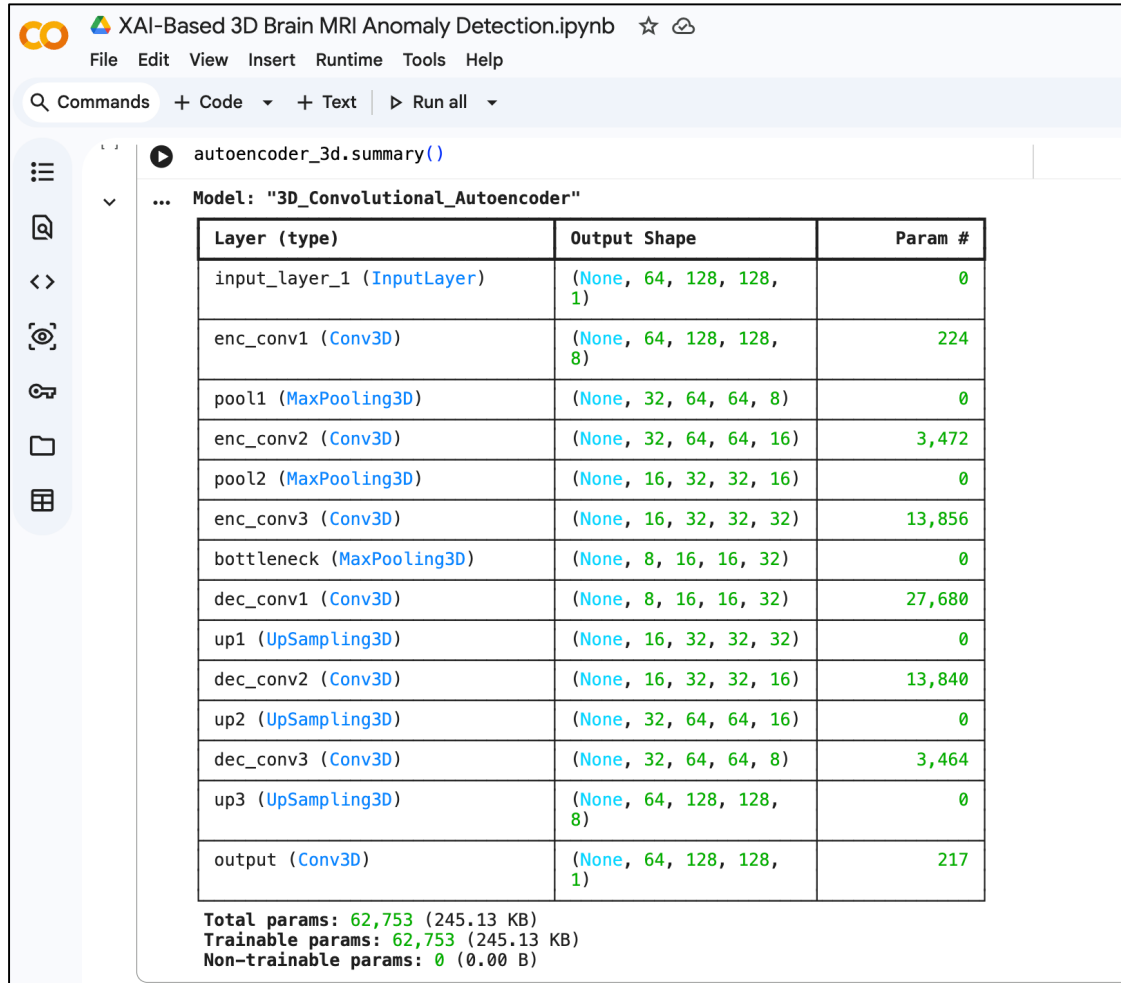


Figure 5.4(b) Keras model summary of the implemented 3D convolutional autoencoder.

After training, the model was used to reconstruct unseen Healthy, MCI, and AD MRI volumes. The reconstruction behaviour was then used for anomaly-score calculation and final performance evaluation.

5.5 Training Configuration

The model is trained using the Adam optimizer and Mean Squared Error loss function. Mean Squared Error is used because the task is reconstruction-based, and the objective is to reduce the difference between the original input and reconstructed output. The main training settings are shown in Table 5.5

Parameter	Value
Input size	$64 \times 128 \times 128 \times 1$
Training data	Healthy MRI volumes only
Optimizer	Adam
Learning rate	0.0001
Loss function	Mean Squared Error
Batch size	1
Epochs	20
Validation data	Healthy MRI volumes only

Table 5.5 Hyperparameter settings used for training the proposed 3D convolutional autoencoder.

During training, both training loss and validation loss are monitored. A decreasing and stable loss pattern indicates that the model is learning to reconstruct healthy MRI samples properly.

5.6 Threshold Selection

After training, the autoencoder is applied to healthy validation samples to compute reconstruction errors. Those validation errors are used to define the threshold of abnormality. The threshold was selected as the **95th percentile** of the reconstruction errors obtained from the healthy validation set. This means that most healthy validation samples should be below the threshold. During the testing, if the reconstruction error of the sample is larger than the threshold, the sample is classified as abnormal. Otherwise, it is normal. This thresholding approach maintains the system as unsupervised, because no MCI or AD labels are used to select the threshold.

5.7 Evaluation Metrics

The model is evaluated as a binary anomaly detector. Healthy samples are treated as normal, while MCI and AD samples are treated as abnormal during testing.

The following evaluation metrics are used:

1. **Accuracy:** Measures the overall correct predictions.
2. **Precision:** Measures how many predicted abnormal samples are actually abnormal.
3. **Recall / Sensitivity:** Measures how many actual abnormal samples are correctly detected.
4. **Specificity:** Measures how many healthy samples are correctly identified as normal.

5. **F1-score:** Provides a balance between precision and recall.
6. **ROC Curve:** Shows the model's ability to separate normal and abnormal samples at different thresholds.
7. **AUC:** Measures the overall separability between normal and abnormal MRI samples.
8. **Confusion Matrix:** Shows true normal, true abnormal, false positive, and false negative predictions.

5.8 XAI Output Generation

To make the model output more interpretable, visual explanation results are generated after testing. The XAI outputs are used to understand which regions contribute to the abnormality decision.

The generated outputs include:

1. Original MRI image or slice
2. Reconstructed MRI image or slice
3. Reconstruction difference map
4. Grad-CAM-based heatmap
5. Heatmap overlay on the MRI image

The reconstruction difference map highlights the areas where the model fails to reconstruct the input accurately. Grad-CAM-based heatmaps provide attention-based visualization of regions that influence the anomaly response. These outputs make the system more transparent and useful for clinical interpretation.

CHAPTER 6

RESULTS

6.1 Training and Validation Loss

The training and validation loss curves are used to analyze how well the autoencoder learns to reconstruct healthy MRI samples. Since the model is trained only on healthy data, the loss value represents the reconstruction difference between the original MRI input and the reconstructed output.

During training, the loss is expected to decrease gradually as the model learns normal brain patterns. As shown in Figure 6.1 a stable validation loss indicates that the model can reconstruct unseen healthy samples without severe overfitting.

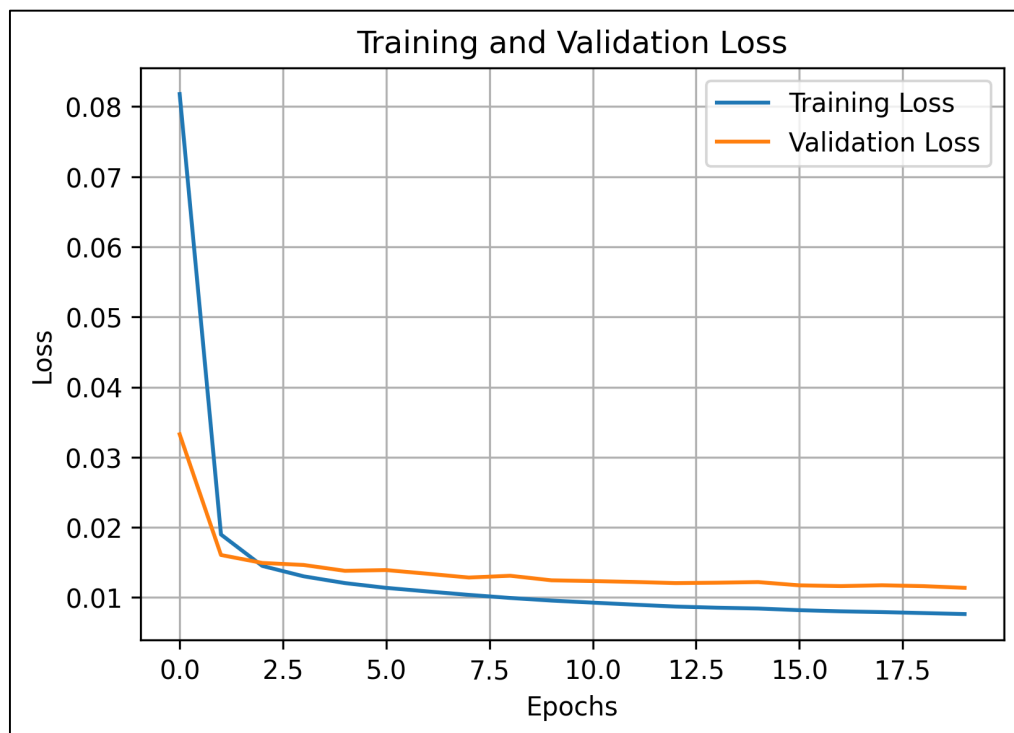


Figure 6.1 Training and validation loss curves during autoencoder training.

The loss curve shows that the model learns useful representations from healthy MRI samples. When the training and validation losses follow a similar decreasing trend, it indicates that the model is learning normal anatomical patterns in a stable manner.

6.2 Anomaly Score Distribution

After training, an anomaly score is calculated for each test MRI sample. Healthy samples are expected to follow the learned healthy reconstruction pattern, while MCI and AD samples may show reconstruction behaviour that deviates from the healthy validation pattern. The anomaly score distribution is analyzed across the following groups as shown in Figure 6.2.

1. Healthy / Non Demented
2. MCI / Very Mild Demented
3. AD / Mild Demented and Moderate Demented

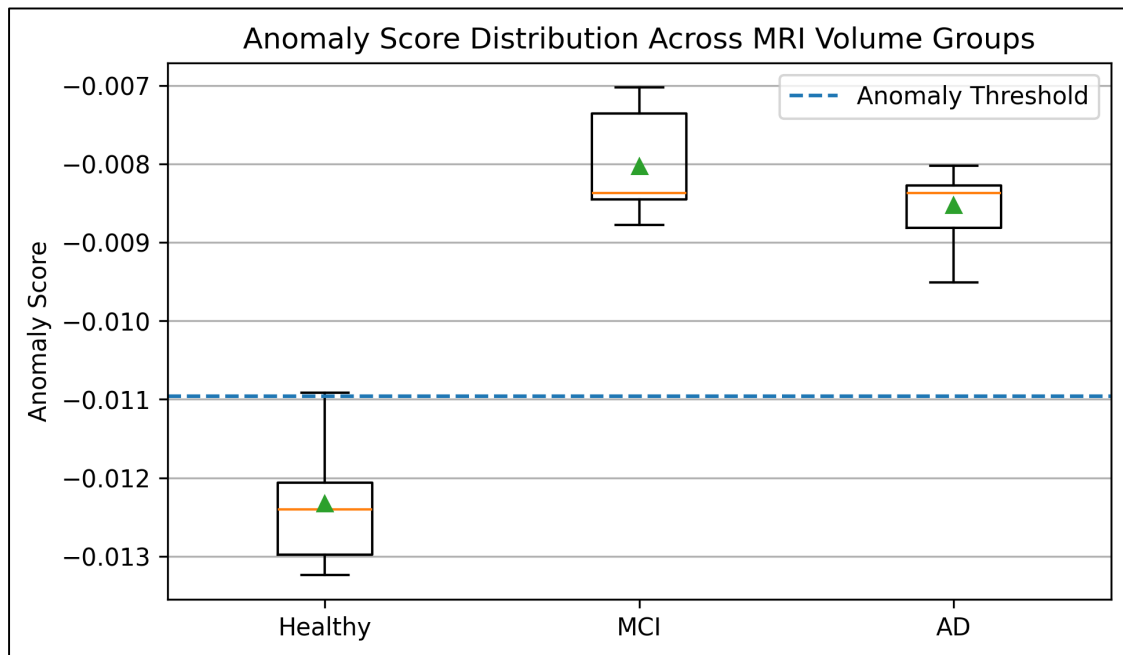


Figure 6.2 Anomaly score distribution across Healthy, MCI, and AD MRI volume groups.

The obtained distribution shows that healthy samples are separated from MCI and AD samples based on their anomaly scores. This indicates that the proposed reconstruction-based framework can use anomaly-score behaviour to support normal and abnormal MRI volume detection.

6.3 Threshold-Based Detection Results

A threshold is selected using reconstruction information obtained from healthy validation samples. In this implementation, the threshold was calculated from the healthy validation reconstruction-error distribution, and the obtained threshold value was 0.01185692. This

threshold is then used to separate test MRI volumes into normal and abnormal categories as shown in Figure 6.3. The decision rule is:

Normal: if the anomaly score is within the healthy range

Abnormal: if the anomaly score crosses the selected threshold

This method keeps the detection process unsupervised because the threshold is derived only from healthy validation data. MCI and AD labels are not used during training or threshold selection. They are used only during evaluation to verify how well the model separates normal and abnormal MRI volumes.

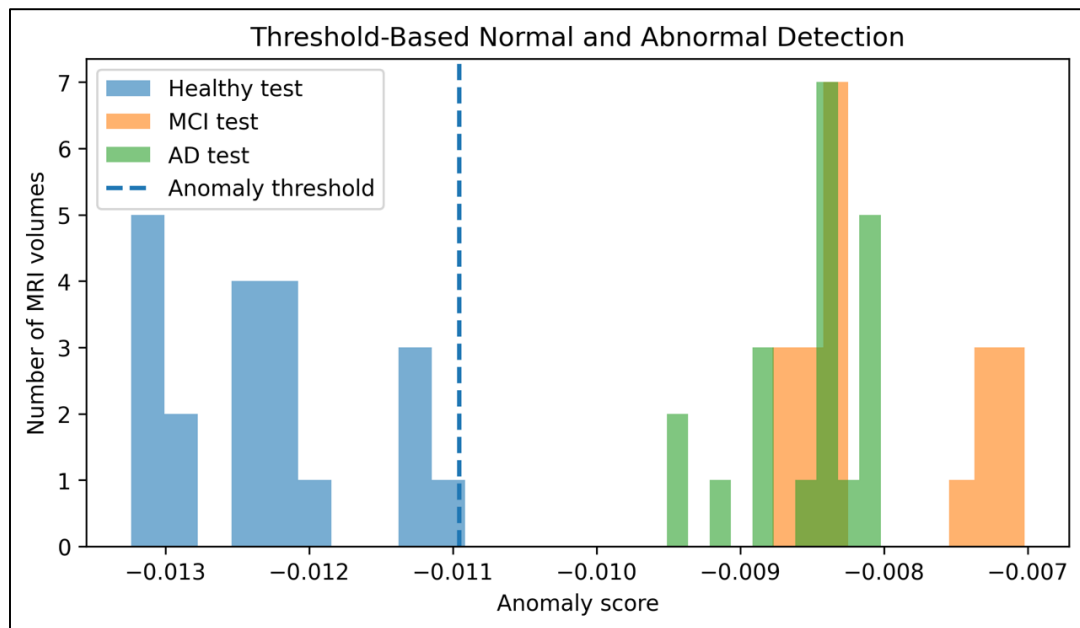


Figure 6.3 Threshold-based normal and abnormal detection result.

The threshold-based output helps determine whether an unseen MRI volume follows the learned healthy reconstruction pattern or deviates from it. In the obtained result, healthy volumes are mostly located on the normal side of the threshold, while MCI and AD volumes are separated toward the abnormal region. This shows that the selected threshold supports normal and abnormal decision-making for the evaluated test set.

6.4 Quantitative Performance Evaluation

The quantitative results in Table 6.4 show that the proposed framework achieved strong performance across all three evaluation scenarios. Both Healthy vs MCI and Healthy vs AD produced high accuracy, sensitivity, specificity, F1-score, and AUC values. The combined Healthy vs MCI + AD scenario also showed strong performance, indicating that

the anomaly-score-based framework can separate healthy and abnormal MRI volumes effectively in the evaluated test set.

Evaluation Scenario	Accuracy (%)	Precision (%)	Recall / Sensitivity (%)	Specificity (%)	F1-score (%)	AUC
Healthy vs MCI	97.50	95.24	100.00	95.00	97.56	1.00
Healthy vs AD	97.50	95.24	100.00	95.00	97.56	1.00
Healthy vs MCI + AD	98.33	97.56	100.00	95.00	98.77	1.00

Table 6.4 *Quantitative evaluation of the proposed 3D convolutional autoencoder using Healthy vs MCI, Healthy vs AD, and Healthy vs MCI + AD benchmark scenarios.*

6.5 ROC Curve and Confusion Matrix

The ROC curve shown in Figure 6.5(a) is used to analyse the separation between normal and abnormal samples at different threshold values. A higher AUC value indicates better separation between healthy and abnormal MRI samples.

The confusion matrix as shown in Figure 6.5(b) provides a summary of correct and incorrect predictions. It shows how many healthy samples are correctly identified as normal, how many abnormal samples are correctly detected, and how many false positive or false negative predictions occur.

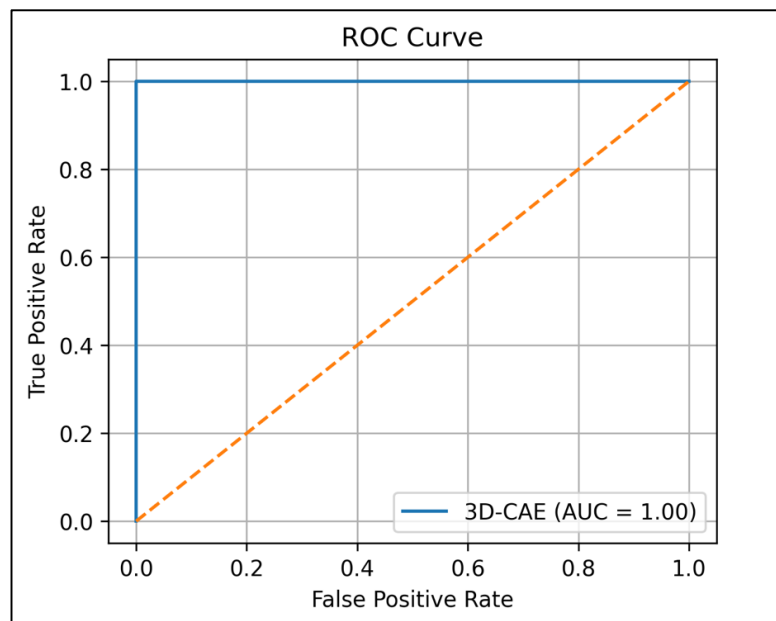


Figure 6.5(a) *ROC curve for neurodegenerative anomaly detection.*

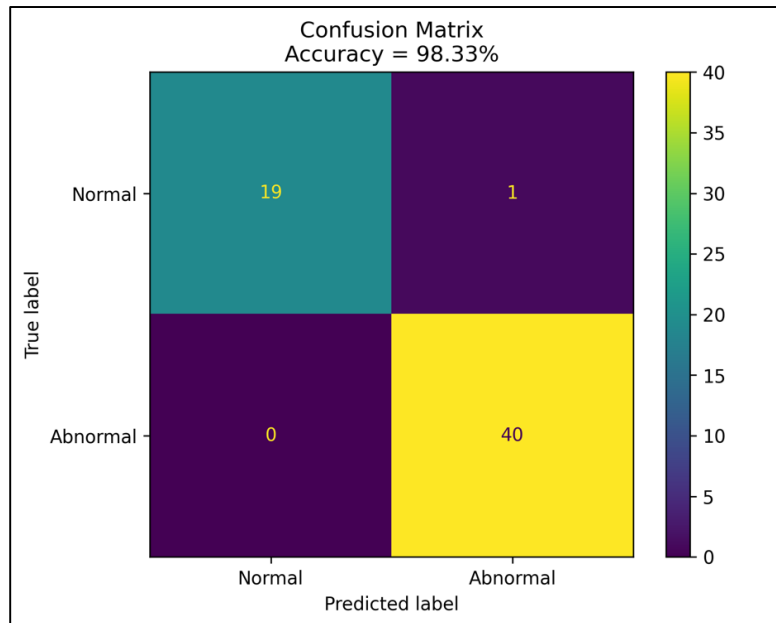


Figure 6.5(b) Confusion matrix for neurodegenerative anomaly detection.

A high sensitivity value is important because the system should detect abnormal neurodegenerative cases effectively. Specificity is also important because too many false positives may reduce the practical usefulness of the system.

6.6 XAI Visualization Results

In addition to numerical performance evaluation, the proposed system generates visual explanation outputs, as shown in Figures 6.6(a), 6.6(b), and 6.6(c). These outputs help interpret why a sample is identified as normal or abnormal.

The XAI visualization results include:

1. Original MRI image or slice
2. Reconstructed MRI image or slice
3. Reconstruction difference map
4. Grad-CAM heatmap
5. Grad-CAM overlay

Healthy MRI samples generally show comparatively lower reconstruction differences, while MCI and AD samples may show stronger difference regions and attention patterns. These visual outputs make the system more interpretable because the model does not only give a numerical score but also shows the regions contributing to the abnormality decision.

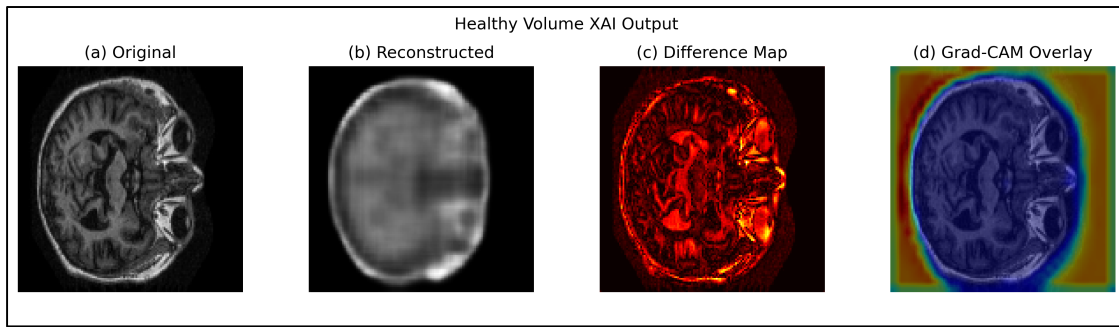


Figure 6.6(a) Representative XAI visualization results for Healthy MRI samples.

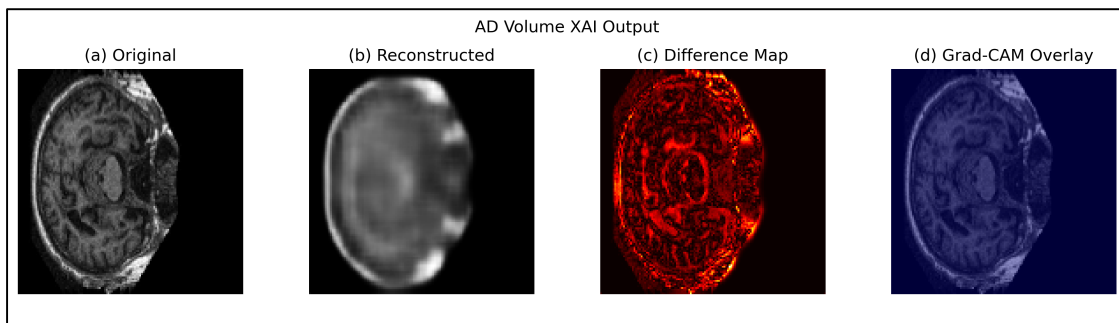


Figure 6.6(b) Representative XAI visualization results for AD MRI samples.

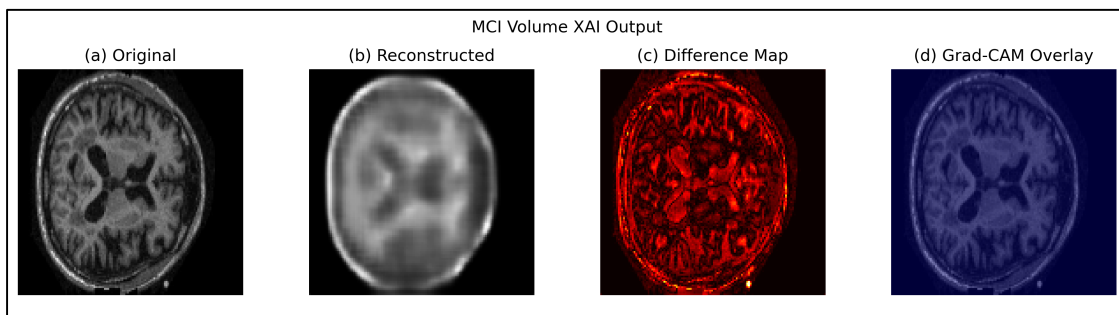


Figure 6.6(c) Representative XAI visualization results for MCI samples.

CONCLUSION

This paper presented an XAI-based 3D unsupervised framework for early neurodegenerative anomaly detection from brain MRI data. The primary goal of the framework is to identify abnormal patterns in the brain without utilizing disease specific labels during training phase. For that the model is trained only on healthy MRI samples, so that it can learn the normal anatomical structure of the brain. An autoencoder based reconstruction approach is used in the proposed system. During testing, the trained model reconstructs the MRI samples and the reconstruction error is used as the anomaly score. A threshold is then calculated from the healthy validation samples, and used to classify the test samples into normal or abnormal. The framework is thus applicable for disease-agnostic anomaly detection where MCI and AD samples are only considered anomalous at evaluation time. The results indicate that reconstruction error can be a useful indicator of neurodegenerative abnormalities. The results show strong performance across Healthy vs MCI, Healthy vs AD, and combined Healthy vs MCI + AD scenarios. This indicates that the proposed reconstruction-based anomaly detection framework can separate healthy and abnormal MRI volumes effectively in the evaluated experimental setup. The system under consideration is proposed as a binary anomaly detector and is assessed with the aid of ROC curve, confusion matrix and quantitative metrics. In terms of interpretability, the system provides XAI-based visual outputs, which include maps of the difference of reconstruction and Grad-CAM based heatmaps. These outputs aid in visualizing the model and help in identifying regions on which the model bases its decision of abnormality, providing a level of transparency for clinical understanding. Thus the proposed approach is not only offering numerical results but also supporting visual explanation. In summary, the proposed framework offers a scalable and interpretable way forward for AI-assisted neuroimaging screening. It can enable early abnormality detection from brain MRI data and can be extended in future with larger official OASIS and ADNI datasets, improved 3D architectures and additional clinical validation.